

the specific ^{di rectory} sudo - Privileges extra
 specified ^{comand} find ip address of machine
 if computer ifconfig
 file system ip address show
 id Change file permission
 -f Linux Architecture
 ps aux Software

loop through list of item
 debug
 set -x

Cloud fundamentals

cloud computing -

Types of services

IAAS
 PAAS
 SAAS ^{private}

Public / Cloud / Hybrid

Availability Region

↓
 collection of
 availability zone

Horizontal Scaling

nodes $\rightarrow$
 Scale in & scale out

Vertical Scaling

CPU & Ram
 Scale up & Scale out

Daemon in Linux

Background Services
 in boot time

\$@ & \$*
 10 ↓ "1" "2" "3" "4"
 "1" "2" "3" "4"
 Separate

Pass arguments to shell

-read
 -read -p

Check if file existing in
 shell

find if (-)
 eg " "

tail in linux
 Bottom 10

Serverless computing

Provider → dynamically resource scalability purpose

It does depend on server for

Resource provisioning
infrastructure
& management

Machine azure

Event triggered

load balancer

① Cost Management
tool

① Pricing - to calculate price of resource we are going to use

② TCO - where

on premises to transfer cloud cost

Azure DevOps

Components

- Azure Board
- Azure Repo
- Azure Test Plan
- Azure Artifacts
- Azure Pipeline

Azure Repository

- Provisioned
- to store repo

owning Github → Microsoft
DevOps Organisation → organisation

CI/CD

CI = Develops code merge

CD = deployment →

Benefits of CI/CD

- Deployment faster
- Reduce error

Blue Green Deployment

Switch b/w current version

Amazon
older

to new version
New one

switch
directing the traffic

ensure pipeline is secure

- variable
- secrets
- RBAC

container
HCM, pipeline

git rebase

a y rep
b
c }

classmate

Date _____

Page _____

add top

on the branch

git merge base

add

→ finding the best
base commit to merge

it.

git tags ← tag provides
commit

for give tag

git tag any repo

tag - d → delete

tag - u → update

git pull & get fetch

↓
pulling repository
in the system
& merges.



push repo
doesn't merge

git merge - 2 Branch

↓
merge

development list together
commits

git stash → temporary
storage save &
go to other branch

git clone
own repository

git fork

someone repository

↓
fetch in our

own repository

Purpose
of terraform

Initialise
terraform
project

terraform
init

• git ignore

- Not want to commit
- Not track

Merge Conflict

& how to resolve

which one to go line
developer.

providers in terraform
template

Provider block

Secure terraform
state file

→ cloud pe
raph
ke secure
karr lena
store current
infrastructure
state

everytime u do
changes state file
get modified.

Purpose of terraform

st

terraform destroy

cli →

terraform workspace

is a component
manage different

env, like

test, prod, stage

Creating workspace

container & image

container

Date _____

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CMD & Entry
Point

CMD - default argument
entry - container
point executable

Docker volume

Store

docker - v

Stop & re

docker stop con

docker contain

docker run →

docker exec -it → runs
existing
containers

Master Node in Kub
manages cluster

Purpose of kubectl

CLI to execute
task in Kubernetes

How to manage scaling
in Kubernetes

⇒ replicas set

kubectl scale --replica

Services in

service attribute in
Kubernetes

⇒ to access our
Pods

Namespace - used to
create different
environment where
+ app

How do I edit
Pod configuration

→ kubectl edit

delete pod

kubectl delete pod podname

SRE

Primp

scalability, reliability
& availability → ensure
site reliability by engineers

SLA - Ser b/w

Agreement Providers
& Client type achieves
this much

SLO - Service Level Objective

They make the target
so that they have
to reach that
target

SLI - service level in
which target is reached

Abbreviation of
SLI, SLA,
SLO

What is error budget
Margin of error we can
handle

Common tools SRE

- Splunk
- Prometheus
- Grafana

SRE & Traditional IT Operations

Engineering solution high
reliability
automate, reduce toil

Store output in a variable

Assign value

redirect → output
store

DevOps Organisation

Create multiple
project & repositories

outside

debug
y
→ bash -x file.chan

classmate

-x file.chan

Page

Cloud computing

Region → collection
of AZ

AZ → data center

DevOps Organi

Service Connection
in Azure DevOps
Credentials

Pipelines connect

YAML - Yaml Ain't Markup
language

key-value pair → indentation

CI/CD

CICD tools

Jenkins, GitLab

Active running - Blue

New version - Green
about to deploy

Commit history

→ git log

Sequence

rebase

reapply commits
on top of branch

Revert specific commit
in git

git revert

git tags - tag

Remember

Primary use of terraform

Terraform

ensure terraform state
file is secured or
not

What is terraform
state

current state → ^{changes} then
it automatically
get updated.

terraform module
reusable block

types of module

① Root module ^{main config file store}

② Child module

↓

Purpose of +
what we have
create destroy →

auto debing
terraform
console

Image →

containers running
instance of image

How u create docker
file

docker build

Share a docker
image

docker
registry / Docker
hub

Inspect

docker logs

or

docker inspect container id

stop & remove all container

docker rm container

docker
run
create
container

docker exec

Go inside
container &
execute
commands
there

Master Node

classmate

Date

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kubectl →

Pods — Smallest +
running objects

scale → Replica Set

Persistent data in kubernetes

Persistent volume,
PVC → Persistent

Namespace in kubernetes
Separated —

SRE

Primary goal

SLI - Metric form

SLO - Objective like
available

→ margin

Error budgeting -
represent

maximum reliable time
1% → stall rate hai